

ASSIGNMENT-2 Maincrafts

1. Introduction

This report documents the methodology, results, and conclusions from Task 2 of the Maincrafts Technology AI/ML internship. The objective was to train and compare three machine learning models — Linear Regression, Ridge Regression, and Decision Tree — on the California Housing dataset to predict median house values (MedHouseVal).

Building upon Task 1 (which trained only a Linear Regression model), Task 2 introduces:

- Feature scaling using StandardScaler to bring all input features to the same scale
- Training of three separate models to identify the best performer
- Comparative analysis using RMSE and R² Score metrics
- Visualisation of model performance, prediction accuracy, and feature importance

2. Methodology

2.1 Dataset

The California Housing dataset was loaded directly from scikit-learn using `fetch_california_housing(as_frame=True)`. It contains 20,640 samples and 9 features describing block-level housing attributes across California.

Feature	Meaning
MedInc	Median income in the area (~\$83k scale)
HouseAge	Average age of houses in the block
AveRooms	Average rooms per household
AveBedrms	Average bedrooms per household
Population	Population of the block
AveOccup	Average number of people per household
Latitude	Geographical latitude coordinate
Longitude	Geographical longitude coordinate
MedHouseVal	Median house value — TARGET variable (capped at 5.0)

2.2 Data Preprocessing

Key preprocessing steps applied before model training:

- **Feature-Target Split:** The target column `MedHouseVal` was separated from the feature matrix `X` using `df.drop()`.
- **Feature Scaling:** `StandardScaler` was applied to `X` (fit on training data only, then transformed on both train and test sets). This prevented features like `Population` (mean ~1425) from dominating over `AveBedrms` (mean ~1.1) purely due to scale differences.
- **Train/Test Split:** Data was divided 80% training (16,512 rows) / 20% testing (4,128 rows) using `train_test_split()` with `random_state=42` to ensure reproducibility across runs.

Note on the California Housing Dataset — Known Data Issues:

- `AveRooms` and `AveBedrms` contain extreme outliers (e.g., 141.9 rooms per household) caused by blocks with very few households but many total rooms.
- `AveOccup` has unrealistically high values (up to 1,243 people per household) — a known data artefact.
- `MedHouseVal` is hard-capped at 5.0, meaning all houses priced above this threshold are recorded as 5.0. This limits the model's ability to differentiate high-value homes.
- No missing values were found in the dataset.

2.3 Models Trained

Three regression models were trained and evaluated:

Model	Description	Key Parameter
Linear Regression	Fits a straight line through the data. Simple and fast; used as baseline.	Default settings
Ridge Regression	Like Linear Regression but adds an L2 penalty to prevent over-reliance on any one feature (overfitting). More stable when features are correlated.	<code>alpha = 1.0</code>
Decision Tree	Builds a flowchart of yes/no decisions to arrive at predictions. Captures non-linear relationships between features.	<code>max_depth = 5</code>

2.4 Evaluation Metrics

Each model was evaluated using two metrics:

- **RMSE (Root Mean Squared Error):** Square root of the average squared prediction error. Penalises large mistakes more heavily. Lower is better.
- **R² Score (Coefficient of Determination):** Measures what proportion of variance in house prices the model explains. Range 0–1 where 1.0 is a perfect fit and 0 means the model does no better than always predicting the mean price.

3. Results

3.1 Model Performance Comparison

The table below summarises the performance of all three models on the test set (4,128 samples):

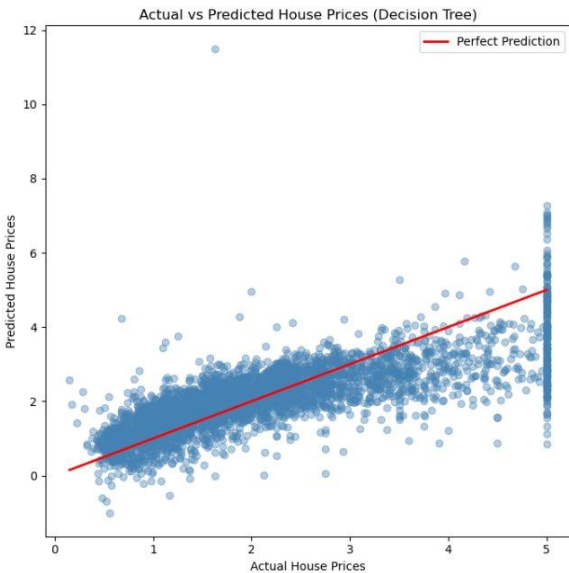
Model	MAE	RMSE	R ² Score
Linear Regression	0.533	0.7456	0.5758
Ridge Regression	—	0.7456	0.5758
Decision Tree	—	0.7242	0.5997

Key Finding: The Decision Tree achieved the best overall performance with the lowest RMSE (0.7242) and highest R² Score (0.5997), explaining approximately 60% of the variance in house prices.

Ridge Regression produced identical results to plain Linear Regression, suggesting multicollinearity was not a significant issue in this dataset.

3.2 Plot 1 — Actual vs Predicted House Prices (Decision Tree)

The scatter plot below compares actual house prices (x-axis) against predicted prices (y-axis) for the Decision Tree model. The red line represents the perfect prediction line ($y = x$) — points closer to the line indicate more accurate predictions.

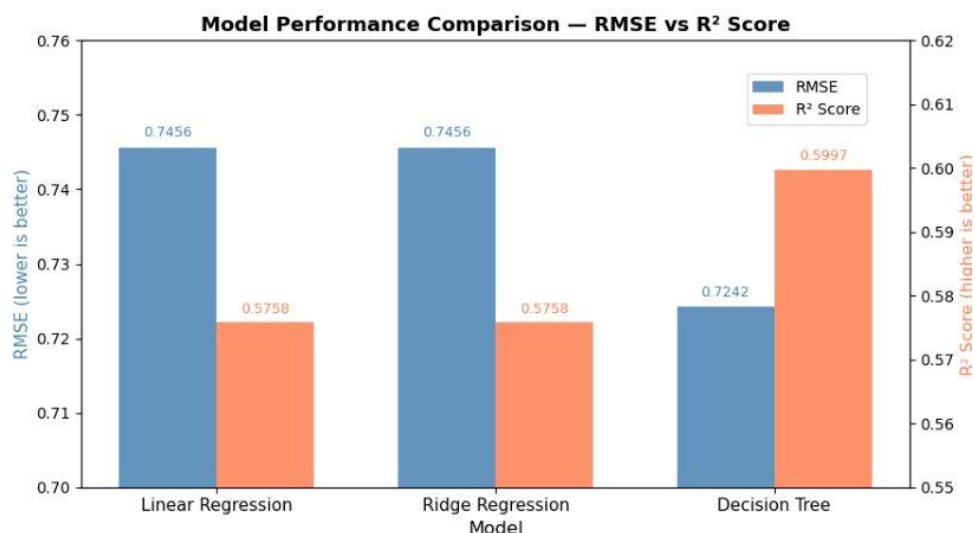


Observations from the plot:

- The predictions broadly follow the positive trend of the perfect prediction line, confirming that the model learned meaningful patterns from the data.
- Predictions cluster tightly around the red line in the mid-price range (1.0 – 3.5), indicating the model performs best for average-priced homes.
- At the upper price boundary (actual = 5.0), predictions show significant spread — a vertical band of points — caused by the MedHouseVal cap in the dataset. Many genuinely expensive homes are recorded as 5.0, but their true values differ, so the model struggles to predict them accurately.
- A few outlier points sit far above the red line (predicted prices ~11–12 for low actual prices), likely caused by blocks with anomalously high AveRooms or AveBedrms outlier values discussed in the preprocessing section.

3.3 Plot 2 — Model Performance Comparison (RMSE vs R^2 Score)

The grouped bar chart below visualises the RMSE (blue, left axis) and R^2 Score (orange, right axis) for all three models side by side.

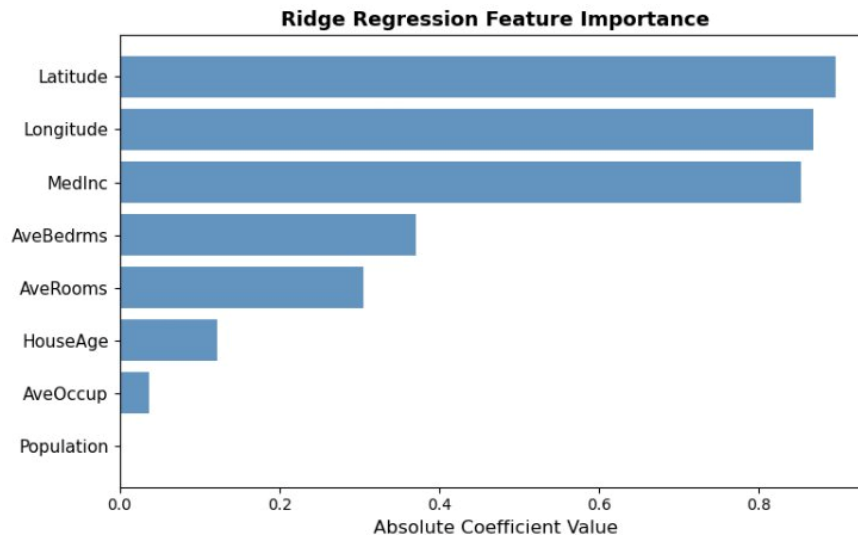


Observations from the chart:

- Linear Regression and Ridge Regression are identical in both metrics (RMSE = 0.7456, R^2 = 0.5758), confirming that the Ridge penalty provided no benefit — the features in this dataset are not strongly enough correlated to cause the overfitting that Ridge is designed to prevent.
- The Decision Tree outperforms both linear models, achieving a lower RMSE (0.7242) and a higher R^2 Score (0.5997). This improvement, while modest (~2.9% better RMSE), demonstrates that house prices have some non-linear patterns that a linear model cannot capture.
- All three models cluster within a narrow performance band, suggesting the ceiling for simple models on this dataset has been reached. Ensemble methods such as Random Forest or XGBoost are expected to yield significantly better results ($R^2 > 0.80+$).

3.4 Plot 3 — Ridge Regression Feature Importance

The horizontal bar chart below shows the absolute coefficient values from the Ridge Regression model, representing how much influence each feature has on the predicted house price.



Observations from the feature importance chart:

- Latitude and Longitude are the most influential features, with absolute coefficients above 0.85. This highlights that geographical location is the single strongest driver of house prices in California — a well-known real-world phenomenon where coastal and urban areas command significantly higher values.
- MedInc (Median Income) ranks third with a coefficient near 0.85. This aligns with the feature correlation heatmap from Task 1, which showed a correlation of 0.69 between MedInc and MedHouseVal — the strongest direct correlation with the target.
- AveBedrms and AveRooms have moderate importance (~0.38 and ~0.30 respectively). Despite these two features being highly correlated with each other (0.85 from the Task 1 heatmap), Ridge Regression successfully distributes the weight between them rather than allowing one to dominate.
- HouseAge contributes minimally (~0.12), suggesting that the age of a house is a relatively weak predictor of its market value once location and income are accounted for.
- AveOccup (~0.03) and Population (~0.00) have near-zero importance, confirming that occupancy density and population size in a block contribute very little to predicting house prices.

4. Conclusions

4.1 Summary of Findings

Task 2 extended the foundational work from Task 1 by introducing feature scaling, training three distinct models, and conducting a structured performance comparison. The key findings are:

- The Decision Tree was the best performing model (RMSE = 0.7242, R^2 = 0.5997), beating both linear models, though by a narrow margin.
- Ridge Regression offered no advantage over plain Linear Regression on this dataset, as multicollinearity between features was not severe enough to cause instability.
- Geographical location (Latitude, Longitude) and economic status (Median Income) are by far the most important predictors of house prices in California.
- All three models explain approximately 57–60% of the variance in house prices. The remaining 40% is likely attributable to features not in the dataset (school quality, proximity to amenities, crime rates) and the artificial ceiling imposed by the MedHouseVal cap at 5.0.

4.2 Key Differences — Task 1 vs Task 2

Aspect	Task 1	Task 2
Feature Scaling	Not applied	StandardScaler applied
Models Trained	Linear Regression only	Linear, Ridge, Decision Tree
Model Selection	Only one model (no comparison)	Best model selected: Decision Tree
Correlation Heatmap	Included (EDA)	Not included
Residuals Histogram	Included	Not included
Model Saving	pickle (.pkl)	joblib (better for large ML models)

4.3 Improvement Recommendations

To achieve higher predictive accuracy in future iterations:

- Try XGBoost or Random Forest: These ensemble methods combine hundreds of decision trees, with each correcting the mistakes of the last. Expected R^2 improvement: 0.60 → 0.83+.
- Engineer smarter features: Instead of raw AveRooms, create derived features like rooms-per-person (AveRooms / AveOccup) which carry more predictive signal.
- Use cross-validation: Replace the single 80/20 split with 5-fold cross-validation for a more reliable estimate of generalisation performance.
- Address the MedHouseVal cap: The hard ceiling at 5.0 prevents the model from learning the full price range. One approach is to train a separate classifier for houses likely above the cap, then handle them independently.
- Handle outliers: Clip or log-transform extreme values in AveRooms, AveBedrms, and AveOccup to prevent these anomalies from distorting model training.

